HostelFlow — End-to-End Walkthrough Guide

Official Step-by-Step Testing & System Demonstration Manual

Ø=Þ€ Ready to Run & Test on Your System!

Frontend Web App: http://localhost:5173 | Backend API: http://localhost:5000

Launch with one click using "start-all.bat" in the root directory.

Part 1: The Student Experience

• **1. Login & Autofill:** Open http://localhost:5173/login. Click the "Student" autofill card (or use student@college.edu / StudentPass123!), then click "Sign In to Dashboard".

• **2. Browse & Apply for a Room:** Click "Browse Rooms" in the sidebar. Filter by sharing tier (Single, Double, Triple, 4-Sharing). Click "Apply For Tier" on your preferred room. Notice the real-time toast confirmation and the status banner indicating your application is pending review.

• **3. Check Maintenance Desk:** Go to "My Complaints" -> Click "Raise New Complaint". Pick an issue category (Electrical, Plumbing, Wi-Fi, Furniture), enter a brief description, attach an optional screenshot, and submit.

Part 2: The Warden Control Room

• **1. Switch to Warden Account:** Click the Logout button in the top navbar. On the login page, click the "Warden Admin" autofill card (warden@hostel.edu / WardenPass123!) and sign in.

• **2. Review & Approve Room Application:** Go to "Applications Queue". Locate the student's pending request and click "Approve". The system executes an atomic transaction (checks capacity -> locks bed -> assigns room -> auto-generates payment reference HMS2026-STUxxxx).

• **3. Inspect Visual Occupancy Grid:** Go to "Overview" (Dashboard). View the live Block & Floor room matrix (Green = Available, Red = Full) updating automatically.

• **4. Triage Maintenance Tickets:** Go to "Complaints Board" (Kanban). Click "Dispatch Technician" to advance ticket to In Progress, then click "Mark Resolved".

Part 3: Completing the External Payment Flow

• **1. Switch Back to Student View:** Logout and log back in as Student. Open "Fee Payment" from the sidebar.

• **2. View Reference & Pay on Smart Portal:** See your newly assigned room and unique Challan Reference ID. Click "Open MMMUT Samarth Fee Portal" (redirects externally to pay).

• **3. Submit Transaction ID & Receipt:** Under Step 2, enter your bank/portal Transaction ID (e.g., TXN98421038), optionally attach an image/PDF receipt, and click "Submit Transaction Details".

• **4. Warden Verification & Room Confirmation:** Switch to Warden -> "Verify Payments". Click "View Proof" to audit the student's details and receipt in the interactive modal. Click "Confirm" to lock the room booking into the Verified state!

HostelFlow — Architecture & Codebase Map

Where Every Component, Model, and Service Lives

1. Codebase Directory Mapping

- **start-all.bat:** Root launcher script. Double-click anytime to launch both backend and frontend and open the browser.
- **client/src/pages/student/:** Student views: StudentDashboard.jsx, BrowseRooms.jsx, PaymentPage.jsx, MyComplaints.jsx.
- **client/src/pages/warden/:** Warden views: WardenDashboard.jsx, ApplicationsQueue.jsx, PaymentVerificationQueue.jsx, ComplaintsManagement.jsx, Analytics.jsx.
- **client/src/components/common/:** Shared UI: Navbar.jsx, Sidebar.jsx, Modal.jsx (proof auditor), StatusBadge.jsx, Loader.jsx, ProtectedRoute.jsx.
- **client/src/api/:** Axios API services: auth.api.js, room.api.js, application.api.js, payment.api.js, complaint.api.js.
- **server/src/controllers/:** Business logic: application.controller.js (atomic transaction lock), payment.controller.js, auth.controller.js, complaint.controller.js.
- **server/src/middleware/:** Security: auth.middleware.js (JWT parsing), role.middleware.js (Role guards), upload.middleware.js (Multer file storage).
- **server/prisma/dev.db:** SQLite database storing all seeded Users, Students, Wardens, Rooms, Applications, Payments, and Complaints.
- **server/uploads/:** Local storage directory storing uploaded receipts and complaint photos.

2. Pre-Seeded Demo Credentials

Warden Admin Account:
Email: warden@hostel.edu | Password: WardenPass123! | Name: Dr. Alok Verma

Student Account:
Email: student@college.edu | Password: StudentPass123! | Name: Rahul Sharma (21BCE1042)

3. Key Project Architecture Decisions

- **MMMUT Samarth Portal Integration:** College portals do not offer third-party payment APIs to student projects. The internal reference challan redirect pattern is the authentic, real-world mechanism for institutional fee reconciliation.
- **Atomic Concurrency Safety:** Room capacity verification and bed counters are executed within Prisma $transaction blocks, completely preventing double-booking race conditions.
- **Audit Trail:** Every administrative decision records timestamps and staff identifiers for complete operational transparency.